

NANO FACTORY NATURAL SHAMPOO

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1 Reception and Storage
of Raw Materials



2 Mixing
Preparation



3 Homogenization



4 Filtration



7 Final product



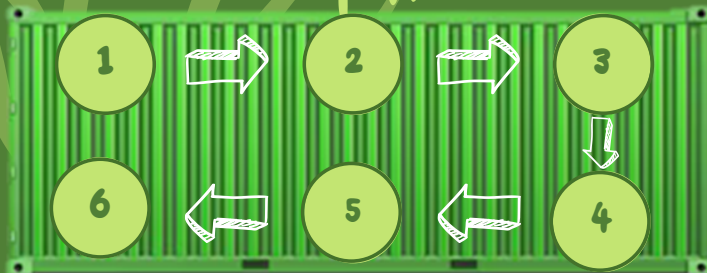
6 Labeling and
Packaging



5 Bottle Filling



Layout



(12 x 2.3 x 2.3) m³

SHAMPOO: RECEPTION AND STORAGE OF RAW MATERIALS

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Subprocess Description:

This subprocess involves the reception and storage of liquid and solid ingredients.

These ingredients are received and stored under specific conditions to ensure the final product's quality.

Additionally, the process includes an automated inventory control system that uses ultrasonic sensors and filling valves to ensure precise and efficient.

Raw Material inputs:

- Water: Main base of the shampoo.
- Surfactants: Responsible for the cleaning action
- Thickeners: Provide the desired texture to the product
- Preservatives: Extend the shelf life of the shampoo
- Fragrances: Provide the characteristic scent to the shampoo.
- Colorants: Give color to the final product.



Flow Diagram:

1. RECEPTION



Arrival of ingredients at the plant.

2 QUALITY CONTROL



Initial quality verification.

3 STORAGE



- Liquids: In temperature-controlled tanks. Solids: In sealed containers.

4 AUTOMATED CONTROL



Ultrasonic sensors in tanks and valves for precise management.

Technical Notes:

-Ultrasonic Sensors: Devices that emit sound waves to measure liquid levels in tanks with high precision.

-Storage Conditions: Liquids are stored at controlled temperatures (between 15°C and 25°C) to prevent ingredient degradation

SHAMPOO: MIXING PREPARATING

Process data

with the information on the left we aim to have a precise weight of each ingredient. We need a weighting scale for this.

After the main ingredients are weighted. Water, Surfactants (Cleansing Agents), Conditioning Agents, Thickeners and Stabilizers, preservatives, fragrances and ph adjusters, are moved to a pot to be mixed.

Automated processes from inbound to outbound
250 L capacity

description

After having received the ingredients in this subprocess each ingredient passes through a weighting scale depending on the formulation of the shampoo



After the scale. These ingredients are mixed with the rest of them depending on the formulation and stored under specific conditions to ensure the final product's quality.



Actuators

- automate out valve
- switch on/off robotic arm
- automated input valve

Sensors

- Weight sensor
- Position sensor

SHAMPOO: HOMOGENIZATION

Process data

- Enters the premix made in the ingredient blending process
- The main ingredients are mixed: water, Surfactants (Cleansing Agents), Conditioning Agents, Thickeners and Stabilizers,
- Preservatives, fragrances and ph adjusters.
- Automated processes from inbound to outbound
 - 250 L capacity



Sensors

- Sensor level
- Temperature sensor
- Mixer speed sensor

Actuators

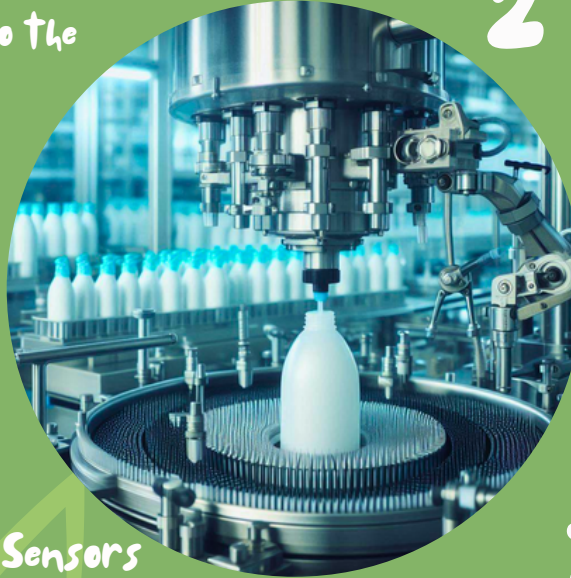
- Atomate out valve
- Switch on/off mixer
- Automated input valve

SHAMPOO: BOTTLE FILLING PROCESS

Subprocess Description:

1 Transport: Bottles are aligned and moved to the filling station.

2 Positioning: Sensors ensure bottles are perfectly aligned under the nozzles.



4 Quality Control: Sensors check fill levels; faulty bottles are removed.

3 Dosage: The machine dispenses the precise amount of shampoo.

Sensors

- Level Sensor: Detects liquid level in the bottle.
- Presence Sensor: Verifies bottle position.
- Optical Sensor: Detects bottle presence.

Actuators

- Pneumatic Cylinder: Moves dispenser for filling.
- Conveyor Belt Motor: Moves bottles.
- Solenoid Valve: Controls liquid flow.

SHAMPOO: FILTRATION

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1. Coarse Filtration:

The raw mixture passes through a coarse filter to remove larger particles and impurities such as undissolved ingredients.

2. Fine Filtration:

Passed through a fine filter with smaller pores to remove finer particles that were not captured in the coarse filtration.

3. Polishing Filtration:

This stage involves passing through an ultra-fine filter or polishing to remove any remaining microscopic particles.

Sensors

- Pressure: Positioned before and after each filter stage.
- Flow Rate: Installed in the pipeline after each filtration stage.
- Temperature: Embedded in the pipeline near the fine and polishing filters.
- Filter Integrity: Attached directly to the filters, particularly in the fine and polishing stages.

Actuator

- Automatic Valve: Connected to the valves controlling the flow into and out of each filtration stage.



SHAMPOO: LABELING AND PACKAGING

Subprocess Description:

Applies pre-printed labels to shampoo bottles, ensuring accurate placement and adhesion.

Seals shampoo bottles with caps, ensuring they are securely closed to protect the product inside.

Wraps groups of shampoo bottles in shrink film, then heats the film so it shrinks tightly around them.

1. Bottle Labeling

2. Bottle Capping

3. Shrink Wrapping



Sensors

- Label Presence Sensor → Detects label availability.
- Bottle Position Sensor → Ensures correct bottle alignment.
- Label Application Verification Sensor → Confirms label is correctly applied.
- Cap Presence Sensor → Detects cap availability.
- Bottle Height Sensor → Checks bottle height for proper capping.
- Cap Application Verification Sensor → Verifies cap is securely applied.
- Film Presence Sensor → Detects shrink film availability.
- Bottle Grouping Sensor → Ensures correct number of bottles for wrapping.
- Temperature Sensor → Monitors heat for proper film shrinking.

actuators

- Label Applicator
- Bottle Positioning Servo
- Label Roll Motor
- Capping Head
- Cap Dispenser
- Bottle Conveyor Motor
- Heat Tunnel Heater
- Film Cutter
- Conveyor Belt Motor

COLOR PALLET

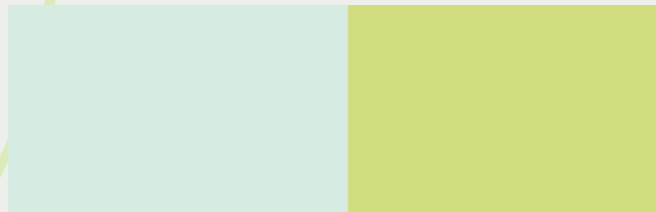


Chelsea Cucumber
#86B665

Yellow Green
#BEE06F

Chalet Green
#4B762F

Fruit Salad
#559859



Swans Down
#D6ECE3

Yellow Green
#D2DE7D